

A Packing of 66 Lines in $\mathbb{R}^{15}$

Limin Ge*

October 8, 2026

Abstract

We give 66 1-dimensional subspaces of $\mathbb{R}^{15}$ whose smallest squared chordal distance is at least 0.924736. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 0.924067, obtained by deleting 1 of the 67 lines of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 66 subspaces of dimension 1 in $\mathbb{R}^{15}$ whose pairwise squared chordal distances $D(a, b) = 1 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 0.924736 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 15$, $n = 1$. The cell $N = 66$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 0.924067, obtained by deleting 1 of the 67 lines of the packing found by this search. The Rankin simplex bound for this cell is $1(15 - 1)/15 \cdot 66/65 = 0.947692$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m15n1N66.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 73 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:
102748541422431429610823376655482800348294901695
denominator:
111111111111151466659381524165299367387316384896

B Coordinates

66 blocks of 15 rows with 1 integers each; divide by 10^{12} . File: `grassmann/packings/m15n1N66.txt`.

518685830800	-120891289453	-195704087932	-19189162520
-373471659068	-450753182119	-23782475455	-107802086735
-136897693689	68159774385	308102251123	274277650285
-92762820704	-136623280305	-234494545881	-403589620262
-129644576429	-507092818282	-473736805266	96892138990
-167064404498	30278559473	320432767916	-355564435052
-304953989850	334041346275	-220892957728	40979184127
-132354132460	-136464098668	398610646154	224304680583
-319608596716	-364854490196	25946047923	-326913393890
349779679898	38049698	-41945088442	-165856967261
155339757696	-302709193438	-17682087207	-386864835917
200784340989	-16344854099	-532246621510	110997927221
-230378916262	-280859740371	488881449930	-196201462439
-247970412664	-1890753001	131145939193	-298314617849
73492744676	313434966971	319580885044	-371805286242
129554420568	-533069797561	103179089796	-312388409529
181077134930	-365982217360	-230702612417	133583428350
62171937396	92943456620	-289610877218	-311804971492
507521989184	-199952984057	-96198118062	-47352640822
291434372562	-2237273476	-89105899686	428211842685
-68362218815	473663353460	-44623559937	233258005540
-344791239032	-51013142159	93443249853	413167529004
415080779186	-210238536189	57579200382	-324089101749
-49319205543	-157508251110	15081679761	309153667844
-253984471953	144009620601	-422919451305	211724451257
-17387070924	-73025619458	-102519909286	129615729534
49943022476	56335829716	-78239566039	-63378008351
90856674757	191889746183	300529064954	-255865744195
-141702656405	21656114888	74461665906	-151932717140
-458400126352	196833232006	374205769848	-147707249946
-223851125721	56559111946	274196773717	-21778709249
-217865607721	-481631764253	35564078993	278930621673
371236187270	524004094526	115065570465	-270897236522
-543994303545	172167753165	44452824255	240611254401
12585990653	243643713238	-483211999800	-51648456627
322027464925	7079438689	-77889322226	18743624566
263571970316	-231480390596	-344017022660	-183702591087
15618420324	-391243465760	-443674223652	279324881679
338499863876	-202759755206	-316488993582	-24533007870
-210821145559	574161297424	29846495343	-215575308240
73163192792	-179704514421	-125307484753	-260203191678
258527116212	-16332418818	74091585717	606013886743
77709679692	-175396656386	332789103121	-17136524669
-188359073753	-389287029525	18205712549	409960724161
-149566562784	123204075470	12062752793	158939673719
72334848658	-239183097021	-65009009945	365226288530
328699369006	234042883504	515306253383	-175114495952
26950240849	276606934968	218462627628	52942911710
-291977606666	-60847717247	277767753721	491086147775
-129504522395	-16613175262	212787203525	129396778914
-405500719603	-227349376303	-145260185554	51097744880
159937193220	-70012398202	342610324330	-14935824412
-233051502959	301719319411	-367735615208	389854101756
178626859133	72325135418	349478486662	147980284059
204980801791	-437855705466	178353385369	472931706992
285855791101	417921776685	320993107276	-118526224222
320830108078	128877060624	-578789769982	282908354627
-175422475260	81946086816	218404865497	163820664454
-473586201870	-370707510346	-27994518072	-22838785046
-161322291060	-251252029205	-184259842632	231095916245
234530836629	-202714225289	94892768703	245917814185
127114880087	-171298406559	-98376265205	529585953556
-275244614084	171560477576	-40349130225	331749233707
-16931094193	-201429825138	287641779769	98540494394
144888020556	-341247052128	-214606460401	365247410701
-15429721865	-206236716346	-210779230412	-43275820042
471971237264	-212153010921	45722608504	15777732963
115284172484	243160476179	-425363328078	224196252113
-81818394618	-190852968723	203290687681	308819919868
-300949736131	301504237629	248286020789	285398379431

69764321418	-30230334663	16834552022	182490207476
-300825929176	-63998595703	-186810861972	384677169114
-70519813018	457046101323	-188390294049	-95306673606
-227106363013	-299206055101	642222238369	76048150452
-11113100903	-100650461631	19457336553	206143182810
-244460433793	143166778984	259731484801	-250931336407
-124818710489	535381023518	279768729291	62002588232
-68843986119	-333886850326	-94040012649	200999661123
248351621193	493908312982	62348327483	70603648906
-165472746358	-172256828382	376541806153	69967014772
504369701834	-164314554910	118604996440	555224300134
296432174345	-132101261398	-271568031754	-225848705465
36353299138	-49586966830	-87522187115	81166363373
147708009831	254200095026	160185112668	-251554703698
-219953091331	370627857036	419175060268	20170361528
-267962404473	176128189346	277064964059	-532976376181
234209558631	387871586179	62997181372	158663172709
-38327788389	107090124956	-294653751349	155500762367
-205824222740	423735984607	-24679576453	121104867747
496578418014	10064227250	559009845449	-432143835208
154093941653	-176174086348	221550911160	125786822842
-60788651788	-239391479118	-207519929483	-8714167450
420804419435	128676802951	-163180300860	-290491259695
197101181774	-162760278552	-281018273457	224811672006
-44243556648	-87584875262	194970729995	113074053612
114687549215	-142757760795	-28392159737	30375152187
-172197794996	176125707542	-83444529429	20061954910
276681321833	-203078500873	19768794748	505013490687
-334265138347	-406709260667	100179619361	211629264404
385926999244	-281351409510	-215797958337	-2677570233
-102038118145	-754636986303	530096551851	-59858232409
-259015348825	210511922907	204840793366	-98215600341
-480409540103	51597929462	-231632378338	65680483400
91949100201	61317270579	-513257705636	-227567895394
-240619362882	-81075897178	153281082796	-560705462925
11772261188	-27500705618	-105820487907	184397258490
195704809858	-14800816715	-64131165738	-126610151813
-175488066912	29376240350	-342097812625	-185370980632
-527494748043	-146966928755	228501287906	-21554108542
375812148305	-45510632109	137405059702	-373221516594
-48360235367	-237920686721	262797601726	360817456223
-197247656459	259380136447	-132565968777	-10983825850
207652476428	44099374873	-417948214057	-213524687875
-260869519897	591933428440	86556464044	464989672950
76213213046	-193476033500	-83757082377	455601068235
260583860266	-110845584080	-308933726126	160446293717
-206086843387	-92625596940	10922927586	134150135032
-249629663449	229041739879	-9733403230	444389942867
-377458536157	-85214146777	-123003887079	-271937871839
194647341123	429487493721	284732806814	134959061127
412647305672	311968498522	449082651468	96699271170
-164281232040	-94189074217	335926982694	-147528208342
105141033111	-302388879731	117314906134	-153879443511
407836198636	-155113966004	-294688649651	265108466981
154407700253	-132756520801	193727642732	330490179556
-488472778534	7729475711	-385648424466	-109181311006
221131712720	73054019369	294148909440	249719261654
-174267177968	-48356379196	290375415390	-176513759899
168756363618	-241698665123	16771327557	-345404083433
76856930223	12973823005	246723647131	75341851870
-140585686197	252951193871	112728793893	-316427200711
5295713463	22423327457	-392204862171	-235597350038
-359411484524	-352942650779	448158474981	-337365737557
-84493742680	-75792889741	182161166067	-456734814685
-305187341998	-68079068682	38785204427	114571808230
349193026966	-386769239996	-212376028361	-93250374473
-242182972723	579740151176	333315186125	368991957177
279185708065	-454829405960	184122000613	-166463356565
-255018176836	218397649322	282421279098	-61194903577
-147020618851	-209247198329	309614072713	204820081510
-36349385658	-348247989317	-14598318481	-489068855773
-132458205601	-419742839366	333212822425	-186790342606
-359472581375	-208053312892	166782816350	71770875630
132986101541	132745526760	-293955171520	-75202562046
-120226353588	-99638236235	353438070864	-537642098014
560478608208	317691090523	279682352544	-4526673324
80876990672	187844753989	204257903860	312911681181
335609704394	80412199240	316266751169	54378295949
181487149918	-403515555867	-27486850257	-418566737338
91335414231	357123977930	-397806968930	-247020529464
317342960635	187854595498	-45872188317	3292649566741
-156125232982	-248009207450	93510313847	-48967204441
687606533841	101893923494	366311610584	-68796533649
88320768109	187210793235	-294733947560	-547864391
43143723823	178883604017	-195465660039	375405781225
-68607943239	261255575893	27257269903	187954893872
306358705003	445621100549	293665817531	267832375049
53546606431	-120449352744	-171508076849	-23272499758
-254928635957	369850381797	162918919773	98760891435
-28555591857	-472938501613	-421980037925	61245869096
166411026996	-377372945345	376481674346	-203185793711
176265281446	-66757568613	122868079477	-444415232194
-70546963760	-267980823116	336587282619	-261691757582
-313906013650	-193518467637	-158424794515	177071872755
250885945048	10010227963	-226631245198	70482446610
86453816734	-81757156499	-29518058075	-121068187328
199874561976	-61107186367	154150483186	73855196026
-48397918861	156535615348	-39215171410	46203347775
160772035333	-329583512744	-442234387953	-242179742240
347705465048	220313767482	291781808648	267991426127
96794031148	119318232047	-162080517083	322878965261
248901807019	175460582282	463807845210	-227597208766

-192919756506	34960974405	-131855448350	259711995982
-553776266488	-125952790094	85733000027	39256601687
192474046163	-202936291565	-275493836841	-308451978894
736925471586	-430438642363	169110580814	-130373227440
155431089218	-108178754910	45933295193	98287779005
50270374933	153448175795	-324295793196	-320553276959
-212707091973	329607921265	-55614569388	-27857628984
-154539376919	-35655614412	125358456029	24705077344
116051048218	235860701023	447973007272	29105047022
-60757031637	-226779622858	-164247159411	15874483930
-267743777570	472585387585	562451954753	-161413127928
-156773533523	69125007440	-360201391635	-161340828354
-211163363226	143970289637	29700873492	737504053334
30809134813	-335070610776	-206568569188	225815640355
-312859675358	321996954350	-165458437758	36189472585
121690023823	201636705905	-29722309371	-185457917224
-228974348608	-205263212939	-236676367813	272881504821
-417364692060	-103038718938	67591720162	150610684048
63716420789	-8353408442	145697533405	355272034976
-8959094912	327785780828	226021108617	-494272872418
-230280487521	-81878779737	354525823391	-437166036861
345124545290	-218203326622	583918917578	28145122209
-23924028724	171159473282	-301920397884	-15494919317
-49614686027	41634190305	-247748097749	130615983185
-178626588053	16282796951	-76078643393	-121844436382
292268387	435103765224	190680287551	-218371701421
7184301571	288564819106	-23971940543	127814935628
367726835182	-135123071715	439019615756	-376175651044
376927819518	-474807679888	102761072022	258920709488
125645354789	-311200905928	31849216001	-238633755564
534268801670	375492062229	-567008820448	-127781202637
-189752374481	225858923721	-106425597871	-217464671591
96675535557	-396177807316	399273325777	-91868082522
66148163330	-451968412435	-119821650342	-394588748047
-297395393686	197595114543	447397684500	-232125594664
195717466432	190408011321	113895275989	351765623363
154956496656	-297820682411	147588937380	-50701702725
-58716200469	83558714950	113218842909	527349336155
397533266039	483636647114	-263989395784	-173270998777
-364655417082	-19960522566	379051761009	-315143992901
75003580110	-358355590604	-51597589611	-85603992054
-325904990494	-24725310166	-19095502596	-240101999211
-270125760591	-27976577068	22165030664	-188887310121
-119189568047	-106600107270	-117126187681	-164352218515
-238476523968	-40852041787	122690381699	-175668905079
448901108492	199070660114	292566204551	-73341543849
289251790625	-101421540317	-43435626173	-68224179776
486829531794	318291588980	-99350409563	52333192545
-38563855471	-208037225964	-69254476279	-180914596465
303978368400	31114747241	139488890452	-184887699439
-367255814049	-364066331630	5773066268	359818423965
-445855986437	34602086886	-106036787156	-411517055517
-67007264753	-386545610947	-247729276348	-572699141664
340978576414	-66547155079	656121297860	348059655739
63047654492	-267899891602	275402626402	-77587590481
129175824393	40790084912	-48954636736	2860411742
173203467941	629760683783	8298423957	-174292448030
-224141678572	133009116046	362220536880	-61680174423
-94285008787	-231588830611	-142279078157	-318274827527
80133458376	77152369756	-381435924728	14736236725
167113393199	-96223115031	81864283077	-485582367206
-265608691364	116908016175	311929714828	-193487470404
220966880437	303930251327	373804296084	179062491772
10314710123	-145922209404	-549293137610	-138483283501
49800991647	-26899612661	176701883160	-246980653162
7665200816	-43112664915	-279220503444	-162710821529
260566755030	526761934372	-8623020872	211830702897
-411021152024	138304736647	189418040628	-89757693984
-267930834763	-227330406695	13032661108	291516714072
-273425846971	-404309199760	118297719584	-391768853807
-286772310885	137340320064	-69030061129	-208394357100
-570014137389	-23360702559	270916751977	244217825682
117103437011	-536999093763	351151192983	-16122568912
298379502755	-154602470945	101624110752	-439449593399
-202358719354	3590881638	288867507574	
114354555738	163220789887	238701532385	